

- This exam contains 9 pages (including the front and back of this cover page) and 15 problems. Check to see if any pages are missing.
- This is the first exam and will count for 25% of your final grade. You have 120 minutes (5:00 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems or multiple choice problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 100 points total in this exam, as well as 1 bonus questions worth a total of 5 possible points.

Page:	3	4	5	6	7	8	9	Total
Points:	22	11	11	20	15	16	5	100
Score:								

Bonus:	
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1. Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

2 pts

- (a) For any set A , $A \subseteq \mathcal{P}(A)$.

☐ True

☐ False

2 pts

- (b) $\neg(p \rightarrow q) \equiv \neg p \rightarrow \neg q$

☐ True

☐ False

2 pts

- (c) If $n \in \mathbb{N}$, then $n - 1 \in \mathbb{N}$.

☐ True

☐ False

2 pts

- (d) If $q \in \mathbb{Q}$, then $\frac{q}{2} \in \mathbb{Q}$.

☐ True

☐ False

2 pts

- (e) The cardinality of the power set of the power set of the power set of the empty set is 3.
That is, $|\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))| = 3$.

☐ True

☐ False

2 pts

- (f) Let $A = \{1, 2, 3, 5, 6, 7, 9\}$ and $B = \{1, 3, 4, 6, 8, 9\}$. Then, $A - B = B - A$.

☐ True

☐ False

2 pts

- (g) Let $A = \{1, 3, 5\} \times \{2, 4, 6\}$ and define $B = \{(s, t) \in A \mid t = s + 1\}$. Then, $|B| = 5$

☐ True

☐ False

2 pts

- (h) For any relation R , $(R^{-1})^{-1} = R$.

☐ True

☐ False

2 pts

- (i) Let S be a relation from A to B , and let R be a relation from B to C . Then $(R \circ S)^{-1} = R^{-1} \circ S^{-1}$.

☐ True

☐ False

2 pts

- (j) $\lg 1 + \lg 2 + \lg 3 + \lg 4 + \lg 5 + \lg 6 = \lg(6!)$

☐ True

☐ False

2 pts

- (k) If R is a transitive relation, then $(a, b) \in R$ implies $(b, a) \in R$.

☐ True

☐ False

4 pts

2. The negation of the converse of $p \rightarrow q$ is:

Ⓐ $q \vee \neg p$

Ⓑ $\neg q \rightarrow \neg p$

Ⓒ $q \wedge \neg p$

Ⓓ $\neg p \rightarrow \neg q$

3 pts

3. $\neg \forall y \exists x Q(x, y)$ is logically equivalent to:

Ⓐ $\exists y \forall x \neg Q(x, y)$

Ⓑ $\exists x \forall y Q(\neg x, y)$

Ⓒ $\forall x \exists y \neg Q(x, y)$

Ⓓ $\forall x \exists y Q(y, x)$

4 pts

4. Determine the power set of $\{\emptyset, 1, \{1\}\}$.

Ⓐ $\{\emptyset, 1, \{\emptyset, 1\}, \{1\}\}$

Ⓑ $\{\emptyset, \{1\}, \{\emptyset\}, \{1\}, \{\emptyset, 1, \{1\}\}\}$

Ⓒ $\{\emptyset, \{1\}, \{\emptyset, 1\}, \{\emptyset, 1, \{1\}\}\}$

Ⓓ $\{\emptyset, \{\emptyset\}, \{1\}, \{\{1\}\}, \{\emptyset, 1\}, \{\emptyset, \{1\}\}, \{1, \{1\}\}, \{\emptyset, 1, \{1\}\}\}$

3 pts

5. Let $S(x, y)$ be the predicate x has seen y where the domain of discourse for x is all students in this class and the domain of discourse for y is all movies. Express the following in logical symbols using quantifiers:

If, for any movie, there is at least one student in the class who has seen that movie, then someone has seen the cinematic masterpiece “*Hellraiser 4: Bloodline*.”

- (A) $\forall y \exists x S(x, y) \rightarrow \exists x S(x, \text{the cinematic masterpiece “Hellraiser 4: Bloodline”})$
(B) $\exists x \forall y S(x, y) \rightarrow \exists x S(x, \text{the cinematic masterpiece “Hellraiser 4: Bloodline”})$
(C) $\forall x \exists y S(x, y) \rightarrow \exists x S(x, \text{the cinematic masterpiece “Hellraiser 4: Bloodline”})$
(D) $\forall y \exists x (S(x, y) \rightarrow S(x, \text{the cinematic masterpiece “Hellraiser 4: Bloodline”}))$

3 pts

6. Let A and B be sets. If A has 10 distinct elements and B has 20 distinct elements, then $|A \times B|$ is

- (A) 10^{20} (B) 200 (C) 30 (D) $(2^{10})^{20}$

5 pts

7. Let R be the relation from $\{1, 2, 3, 4\}$ to $\{l, m, n, o, p\}$ given by $R = \{(1, m), (2, n), (3, p), (1, o)\}$ and let S be the relation from $\{l, m, n, o, p\}$ to $\{a, b, c\}$ given by $S = \{(l, a), (n, c), (p, a), (p, c)\}$. Mark all of the ordered pairs below that belong to $S \circ R$.

- (A) $(3, c)$ (B) $(1, c)$ (C) $(3, a)$ (D) $(2, c)$ (E) $(2, a)$

10 pts

8. Prove the following symbolic argument is valid:

$$\begin{array}{l} p \rightarrow r \\ q \rightarrow r \\ \hline p \vee q \\ \hline \therefore r \end{array}$$

10 pts

9. Give a direct proof that the product of two odd integers is odd.

10 pts

10. Let A be a set and let R be the relation on $\mathcal{P}(A)$ defined by $U R V$ if and only if $V \subseteq U$. Show that R is an order relation—that is, show that R is reflexive, antisymmetric, and transitive.

5 pts

11. Let A and B be sets and let $f : A \rightarrow B$ be a function. Define the relation R on A so that $x R y$ if $f(x) = f(y)$. Is R an equivalence relation on A ? Justify your answer.

5 pts

12. Let A and B be sets, each having a finite number of elements. If $|A| < |B|$, explain why it is not possible to construct a function $f : A \rightarrow B$ that is onto.

13. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{w, x, y, z\}$. Consider the functions $f : A \rightarrow B$ and $g : B \rightarrow A$ defined by

$$f = \{(1, y), (2, x), (3, w), (4, y), (5, z)\}$$

$$g = \{(w, 3), (x, 1), (y, 5), (z, 2)\}.$$

5 pts

- (a) Compute the compositions $f \circ g : B \rightarrow B$ and $g \circ f : A \rightarrow A$.

3 pts

- (b) Mark the properties satisfied by $f \circ g$

☐ (A) one-to-one

☐ (B) onto

☐ (C) bijective

3 pts

- (c) Mark all properties satisfied by $g \circ f$

☐ (A) one-to-one

☐ (B) onto

☐ (C) bijective

5 pts

14. Let $A = \{\pi, \sqrt{2}, e^2, 17.2\}$ and let $B = \{e^2, \sqrt{11}, \sin(e), \pi\}$. Circle all of the true statements in the list below.

☐ (A) $\pi \in A \cap B$

☐ (B) $\pi \in A \cup B$

☐ (C) $\pi \in A - B$

☐ (D) $\sqrt{2} \in A \oplus B$

5 (bonus)

15. $\neg \forall y \exists x Q(x) \rightarrow Q(y)$ is logically equivalent to:

☐ (A) $\exists y \forall x \neg Q(y) \rightarrow \neg Q(x)$

☐ (B) $\exists y \forall x \neg Q(x) \rightarrow \neg Q(y)$

☐ (C) $\exists y \forall x Q(x) \wedge \neg Q(y)$

☐ (D) $\exists x \forall y Q(x) \vee \neg Q(y)$